

Investigation by AI about the mysterious 137

A three-hour, human-steered dialogue between AI systems on the origin of the fine-structure constant

Fable5.1¹, GPT5.6-Sol², l0d0v1c³, and pseudoLuc³

¹AI system (Claude Fable 5.1, Anthropic): calculations, scripts, literature checks, drafting

²AI system: critical review and extension of the first half of the investigation

³Questions, steering, relay between the AI systems, time-keeping

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Abstract

We report an experiment rather than a discovery. In about three hours of conversation, steered by open questions and with one round of cross-criticism between two AI systems, an AI assistant equipped with a Python sandbox and a web-search tool was asked whether the fine-structure constant $\alpha \simeq 1/137.036$ might have a geometric origin. The purpose was to see how far such a dialogue can go, and in particular whether it can *re-derive known results, possibly by other means*, while keeping track of its own errors. The investigation (i) tested 4.8×10^6 products of volumes of symmetric spaces against four constants with a decoy-target control, finding nothing beyond chance and recovering Wyler’s 1969 formula as the unique hit of its class; (ii) showed that Wyler’s value lies *above* the infrared floor of α^{-1} and therefore cannot be α at any scale; (iii) quantified the “induced electrodynamics” hypothesis and scanned 151,263 vector-like spectra, rediscovering the “strong unification” scenario and formulating a *conservation lemma*: a rigid ultraviolet condition trades a modulus for a threshold and removes no continuous parameter; (iv) re-derived at two loops, from independent inputs, that the Standard Model alone unifies for a hypercharge normalisation $k_Y = 1.341 \pm 0.004$, compatible with $k_Y = 4/3$, which then predicts $\alpha^{-1}(0) = 136.7$ (−0.26%); (v) connected $k_Y = 4/3$ to a point of enhanced symmetry in $SU(6)$ and, through a generalised Schellekens condition, to fractional electric charges in units of $1/6$; (vi) stress-tested that conclusion with Kaluza–Klein thresholds, Bayesian evidence, an intermediate supersymmetry scale and the Higgs quartic coupling. Almost every positive result turned out to be already in the literature; we list what was recovered, by which route, what was corrected along the way, and what such a dialogue cannot do. All Python sources are reproduced in the appendix.

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1 Introduction: the experiment

The number $1/137$ has a long record of attracting numerology [1, 2, 3]. It is therefore a good stress test for an AI system: the topic rewards fluent speculation, punishes it quickly when numbers are computed, and has a large literature against which any claim can be checked.

Protocol. The investigation was a single conversation of about three hours (wall-clock time, as kept by the human participants). It started from a naive question, *does α tend to an asymptote at high energy?*, and proceeded by short prompts (“could it come from a geometry?”, “can you try?”, “do you have an idea?”, “go on”). The first AI system (Fable5.1) had access to a sandboxed Linux machine (Python 3 with `numpy` only) and to web search. Midway, the working notes were submitted to a second AI system (GPT5.6-Sol), whose twelve-section critique was pasted back into the conversation; the first system checked every number in it, conceded the points on which it was wrong, objected where it disagreed, and continued. No human supplied physics content; the human role was to ask, to choose which suggested follow-up to pursue, and at one point to ask whether it was worth continuing.

Aims. (a) Recover established results, where possible by a route different from the original one. (b) Attach to every numerical coincidence an estimate of how often chance does as well. (c) Keep an explicit log of mistakes and retractions. (d) State clearly what remains unverified.

What this paper is not. It is not a claim of new physics, and it has not been reviewed by a specialist. The calculations are one- or two-loop estimates with unknown high-scale thresholds. Section 9 lists the known weaknesses, including formulae recalled from the model’s memory.

2 Background: what kind of number is α ?

At one loop, for $Q \gg m_f$,

$$\alpha(Q^2) = \frac{\alpha_0}{1 - \frac{\alpha_0}{3\pi} \sum_f Q_f^2 N_c \ln \frac{Q^2}{m_f^2}}, \quad (1)$$

so that α^{-1} decreases logarithmically with energy, from 137.036 at $Q = 0$ to $\simeq 127.95$ at M_Z , and formally reaches a Landau pole far above the Planck scale. There is no high-energy asymptote. Below the electron mass, however, nothing runs any more: $\alpha^{-1}(0) = 137.035999177$ [5] is an infrared *floor*, the only true asymptote of the curve. Above the electroweak scale the relevant quantities are the hypercharge and $SU(2)$ couplings, with $\alpha_{\text{em}}^{-1} = \alpha_Y^{-1} + \alpha_2^{-1}$.

Two clarifications, both of which emerged from the cross-criticism step, frame everything that follows. First, in the renormalised Standard Model $\alpha(0)$ is an *input*. Fermion masses do not determine it; they only relate the coupling at two scales. Masses become decisive only once an ultraviolet condition $\alpha^{-1}(\Lambda) = C$ is imposed, in which case $\alpha^{-1}(0) = C + \text{running} + \text{thresholds}$. Second, the identity $\alpha = Z_0/2R_K$, with $R_K = h/e^2$ the von Klitzing resistance, is a tautology: the topological nature of the quantum Hall effect fixes the integer ν in $R_H = R_K/\nu$, not the ratio R_K/Z_0 .

3 The direct geometric route, and why it fails

3.1 What topology fixes and what it does not

For a $U(1)$ gauge field,

$$S[A] = \frac{1}{2e^2} \int F \wedge *F + \frac{i\theta}{8\pi^2} \int F \wedge F. \quad (2)$$

Fluxes $\frac{1}{2\pi} \int_\Sigma F$ are integers and θ is periodic, but the Maxwell term contains the Hodge star, hence the metric, and its coefficient $1/e^2$ is continuous. A Chern number explains why charges are integers, not why two unit charges interact with strength $1/137$. A rigid value requires something more: a symmetry (e.g. a fixed point of an $SL(2, \mathbb{Z})$ duality acting on $\tau = \theta/2\pi + i/\alpha$, which gives $\alpha \sim 1$), a dynamical stabilisation, a renormalisation-group fixed point, or a compositeness condition $1/e^2(\Lambda) = 0$.

3.2 Wyler’s formula

Wyler [2] proposed

$$\alpha_W = \frac{9}{8\pi^4} \left(\frac{\pi^5}{2^4 5!} \right)^{1/4}, \quad \alpha_W^{-1} = 137.03608, \quad (3)$$

from Hua’s volumes [4] of the type-IV Cartan domain $D^5 = SO(5, 2)/[SO(5) \times SO(2)]$, its Shilov boundary Q^5 and S^4 . The classical objections (no Lagrangian, arbitrary choice of $SO(5, 2)$ and of the power $1/4$, normalisation-dependent volumes) are due to Robertson [3] and others. The cross-criticism step supplied a sharper one:

$\alpha_W^{-1} = 137.03608$ is *larger* than $\alpha^{-1}(0) = 137.035999$. Since screening implies $\alpha^{-1}(Q^2) < \alpha^{-1}(0)$ for all $Q^2 > 0$, Wyler’s number is on the wrong side of the infrared floor and cannot be reinterpreted as α at any scale.

3.3 Numerical tests with a look-elsewhere control

Wyler’s recipe in dimension n (script 01) gives, for $n = 3, \dots, 8$, the values $\alpha^{-1} = 38.8, 76.9, 137.036, 225.7, 349.8, 515.9$: only $n = 5$ lands near a physical value, and neither 128 nor the unification values 25–47 appear.

Wyler’s class. Among all numbers $2^a 3^b 5^c \pi^d$ with quarter-integer exponents in $[-6, 6]$ (5.8×10^6 combinations), exactly one lies within 10^{-6} of $\alpha^{-1}(0)$: Wyler’s formula, recovered without being put in. None lies within 10^{-7} . The expected number of chance hits at 10^{-6} is 0.1–0.3, so the usual remark that “one can get 137 from anything” is too strong within this class, while the formula is nonetheless excluded at today’s precision.

Four Cartan families (script 02). With 36 building blocks (Hua volumes of domains of types I–IV up to complex dimension 10, spheres S^1 – S^8 , Shilov boundaries Q^2 – Q^8), products of up to three blocks with exponents $\pm\frac{1}{4}, \pm\frac{1}{2}, \pm 1, \pm 2$ and a small rational prefactor give 4.78×10^6 distinct values. For each target, 4000 decoy targets drawn within $\pm 20\%$ estimate the chance rate (Table 1).

Table 1: Hits of the Cartan-volume family (script 02). “Expected” is the mean over 4000 local decoys.

Target	hits at 10^{-5} (expected)	hits at 10^{-6} (expected)	hits at 10^{-7}
α^{-1}	5 (4.1)	2 (0.42)	0
m_μ/m_e	4 (3.9)	1 (0.38)	0
m_p/m_e	3 (3.0)	0 (0.30)	0
m_n/m_e	3 (3.0)	0 (0.31)	0

The two hits for α^{-1} are Wyler’s formula and a meaningless mixture of a type-I domain, a type-III domain and Q^4 that does slightly better (4.4×10^{-7}): once the family is enlarged, Wyler loses uniqueness. The muon hit shares no block with those of α ; $6\pi^5$ for m_p/m_e stays at 2×10^{-5} , within the noise. *The real constants behave like generic numbers with respect to this family.*

4 The indirect route: ultraviolet condition, spectrum, running

If geometry matters, it must act indirectly: it fixes an ultraviolet condition, a spectrum and thresholds, and the renormalisation group produces $\alpha(0)$.

4.1 Induced electrodynamics and the spectral budget

Following Landau and Zeldovich [6, 7], suppose the photon has no kinetic term of its own at a fundamental scale: $\alpha_Y^{-1}(\Lambda) = 0$. With $\Lambda = M_{\text{Pl}} = 1.22 \times 10^{19}$ GeV, $b_Y^{\text{SM}} = 41/6$ and $\ln(M_{\text{Pl}}/M_Z) = 39.44$, the Standard Model induces $\alpha_Y^{-1}(M_Z) = 42.9$ against a measured 98.4; adding the measured $\alpha_2^{-1}(M_Z) = 29.6$ and the 9.1 units gained between M_Z and $Q = 0$ gives $\alpha^{-1}(0) \simeq 82$. The right order of magnitude with no free parameter, but wrong by a factor 1.7. The deficit is a *spectral budget*,

$$\sum_j \Delta b_{Y,j} \ln \frac{\Lambda}{M_j} \simeq 2\pi \times 55.5 \simeq 348.7, \quad (4)$$

which vector-like matter can fill: seven $Y=1$ Dirac leptons at 0.7 TeV, two $Y=2$ fermions at 8×10^4 GeV, or one $Y=3$ fermion at 3×10^6 GeV. These toy solutions (second AI system, checked by the first) show that the pole condition is one *integrated* constraint satisfied by infinitely many spectra. The first system added that Λ is itself a hidden continuous parameter, $M \propto \Lambda^{1+b_{\text{SM}}/\Delta b}$: with the reduced Planck mass the seven leptons drop from 726 GeV to about 45 GeV, excluded since LEP. It also noted that an $SL(2, \mathbb{Z})$ fixed point ($\alpha^{-1}(\Lambda) \simeq 1$) and induced electrodynamics

($\alpha^{-1}(\Lambda) = 0$) differ by 1% of the budget and should be treated as one family, “rigid strong coupling at Λ ”, $\alpha_i^{-1}(\Lambda) \in [0, 1]$, whose intrinsic precision is about 1%.

4.2 Scan of 151,263 spectra (script 03)

We asked which matter contents make all three couplings strong at a common Λ . The family comprises the vector-like representations contained in the **5**, **10** and **24** of $SU(5)$, namely $E(1, 1)_1$, $L(1, 2)_{1/2}$, $D(3, 1)_{1/3}$, $U(3, 1)_{2/3}$, $Q(3, 2)_{1/6}$ with multiplicities 0–6, and a real triplet and octet with multiplicities 0–2, all at a common mass $M \geq 1$ TeV, on a Standard Model base or an MSSM base with superpartners at M . As a control, the three measured couplings were replaced by “decoy worlds” drawn within a factor 1.25–2.

Table 2: Spectra for which (Λ, M) exist with $0 \leq \alpha_i^{-1}(\Lambda) \leq 1$ for $i = Y, 2, 3$ (scripts 03 and 08). Decoy medians are for a spread factor 1.5 unless stated.

Base	Λ	compatible	decoy median	$P(\text{decoy} \geq \text{real})$
SM, one mass	free (10^{10} – 10^{19} GeV)	2565 (1.7%)	1743	0.39–0.49
SM, one mass	Planck window	645 (0.43%)	242	0.013–0.05
MSSM, one mass	Planck window	1110 (0.73%)	688 ($\times 1.25$)	0.025
SM, two masses	free	12,912 (8.5%)	–	–
SM, two masses	Planck window	4941 (3.3%)	2438	0.06–0.31

With Λ free the condition has no explanatory power: thousands of spectra qualify and our world is typical. With Λ in the window $[2.4 \times 10^{18}, 1.22 \times 10^{19}]$ GeV our world looked mildly favoured ($p \simeq 1$ –5%). The cause was identified: at that scale the required Δb_i are nearly equal, (8.84, 7.88, 8.35), which generic matter provides easily; a fictitious world requiring exactly equal Δb_i obtains 639 spectra against 641 for ours, while unbalanced requirements give 266, 44, 8, 0. The preference did not survive a second mass scale (coloured and uncoloured multiplets at different masses, script 08): p rose to 0.06–0.31. We file it as a fluctuation.

A rediscovery. On the MSSM base the only complete- $SU(5)$ -multiplet solutions are $(n_5, n_{10}) = (6, 0), (3, 1), (4, 1), (0, 2), (1, 2)$, i.e. $n_5 + 3n_{10} = 6$ or 7, with $\Lambda \simeq 10^{16}$ GeV and $M \simeq 8$ –80 TeV: the brute-force scan lands, unprompted, on the “strong unification”/“universal Landau pole” scenario [8, 9, 10]. On the Standard Model base no complete multiplet works, as the first system had argued beforehand: equal Δb_i in $SU(5)$ normalisation leave coupling differences intact, so a common pole would require the Standard Model to unify already with $k_Y = 5/3$, which it does not.

4.3 A conservation lemma

In the best structured case (MSSM plus $n = n_5 + 3n_{10}$ multiplets, common strong coupling, Λ fitted to α_s), $\alpha^{-1}(0)$ is predicted (script 04): $n = 6$ gives 126.6–119.9 for $M = 3$ TeV, **139.8–133.1** for $M = 10$ TeV and 151.8–145.2 for $M = 30$ TeV (the range corresponds to $\alpha_i^{-1}(\Lambda) \in [0, 1]$). One can get 137, but

$$\frac{d\alpha^{-1}(0)}{d\ln M} \simeq 11. \quad (5)$$

Lemma (conservation of the free parameter). A rigid ultraviolet condition does not reduce the number of continuous parameters. Ordinary grand unification has (M_U, α_U) ; rigid strong coupling has (Λ, M) . In both cases there are three observables, two parameters and one prediction. The modulus α_U is exchanged for the threshold M , at a rate of 8/3 per unit of α_U^{-1} against 11 per e -fold of M .

Every specialist knows this implicitly; we have not seen it phrased this way, and it recurs below in a second guise (Section 6.3).

The only escape. A prediction requires $\ln(\Lambda/M)$ itself to be discrete. Dimensional transmutation of a hidden gauge sector gives $M = \Lambda \exp[-2\pi\alpha_h^{-1}(\Lambda)/b_h]$. A rigid *strong* coupling cannot generate a large hierarchy (for a confining sector $b_h \gtrsim 5$ gives $\ln(\Lambda/M) \lesssim 1$, and small b_h leads into the conformal window), so the hidden coupling must be *weak*, $\alpha_h^{-1}(\Lambda) = N_h \gg 1$. If N_h is discrete (a flux, a rank), then at leading order $\alpha_i^{-1}(M_Z) = N_i + b_i N_{EW}/b_{EW} + \Delta b_i N_h/b_h$ is an explicit function of integers. Two-loop and threshold corrections amount to a few units, so these integers *cannot* be read off 137.036; we deliberately did not scan for them, since the theoretical error exceeds the spacing of candidates.

5 Where discreteness really enters: the hypercharge normalisation

5.1 A two-loop re-derivation

One discrete, geometric datum does touch α : the normalisation k_Y in $\alpha_1 = k_Y \alpha_Y$. In heterotic string theory $k_i \alpha_{\text{string}}^{-1} = \alpha_i^{-1}$ with k_i the levels of the world-sheet current algebras [17]; this realises the idea “level of a current algebra $\rightarrow$ coupling”, but fixes only *ratios*. The canonical $SU(5)$ value is $5/3$.

We took as inputs the $\overline{\text{MS}}$ couplings at $\mu = M_t$ extracted at NNLO by Buttazzo et al. [13], $\alpha_Y^{-1} = 97.885$, $\alpha_2^{-1} = 29.946$, $\alpha_3^{-1} = 9.262$ (for $\alpha_s(M_Z) = 0.1180$), and ran the two-loop Standard Model equations [14] with the top Yukawa (script 05). Defining M_U by $\alpha_2 = \alpha_3$:

	M_U	α_U^{-1}	required k_Y
one loop	$9.2 \times 10^{16} \text{ GeV}$	47.03	1.297
two loops	$2.9 \times 10^{16} \text{ GeV}$	46.18	1.341 $\pm$ 0.004

The error reflects $\alpha_s(M_Z) = 0.1180 \pm 0.0009$; the top-mass uncertainty is negligible. Fitting (M_U, α_U) to α_2, α_3 and *predicting* α_Y , hence $\alpha^{-1}(0)$, gives Table 3.

Table 3: Predicted $\alpha^{-1}(0)$ for rational k_Y , Standard Model only (script 05). Measured: 137.036.

k_Y	5/3	4/3	13/10	32/25	5/4	6/5	1
one loop	154.4	138.7	137.2	136.2	134.8	132.5	123.1
two loops	152.1	136.68	135.1	134.2	132.8	130.5	121.3
deviation	+11.0%	−0.26%	−1.4%	−2.1%	−3.1%	−4.7%	−11.5%

At one loop the first system had preferred 32/25; two loops moved the requirement from 1.30 to 1.34 and singled out 4/3. The residual 4/3 vs. 1.341 is 2σ of α_s , or 0.36 units of α^{-1} , well below typical unification-scale thresholds. This reproduces Barger, Jiang, Langacker and Li [11, 12], who found excellent two-loop unification of the Standard Model for $k_Y = 4/3$; the route here is independent (different inputs, a prediction of $\alpha^{-1}(0)$ rather than a convergence measure). Figure 1 shows the running.

5.2 Where can $k_Y = 4/3$ come from? (script 06)

Schellekens’ theorem. In weakly coupled heterotic models with $SU(3)$ and $SU(2)$ at level one, the absence of fractionally charged states requires $k_3/3 + k_2/4 + k_Y/4 \in \mathbb{Z}$, hence $k_Y \geq 5/3$ [15, 16, 17]. Any $k_Y < 5/3$ therefore implies fractional electric charges. We implemented the generalised condition of [16]: charges quantised in units of $1/p$ are possible iff $[p \neq 0 \pmod{3}]/3 + [p \neq 0 \pmod{2}]/4 + p^2 k_Y/4 \in \mathbb{Z}$. The minimal p is

k_Y	5/3	4/3	5/4	32/25	13/10
minimal charge unit	1	1/6	1/12	1/30	1/60

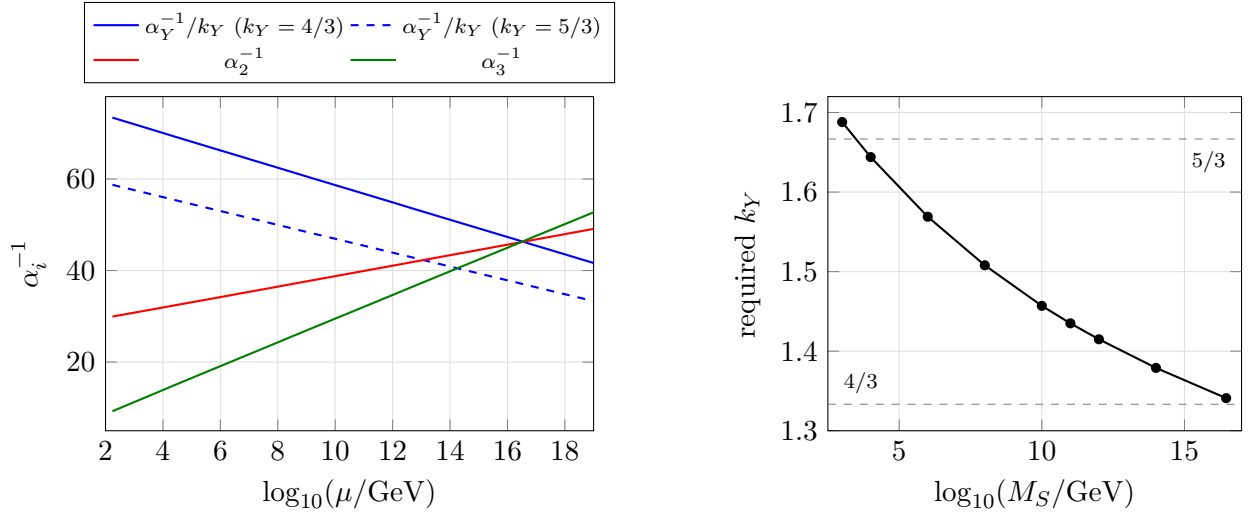


Figure 1: Left: two-loop running of the Standard Model couplings; the hypercharge coupling is shown with the canonical normalisation $k_Y = 5/3$ (dashed) and with $k_Y = 4/3$ (solid). Right: the value of k_Y required for exact unification when the MSSM replaces the Standard Model above a scale M_S ; the last point is the pure Standard Model (script 09).

Within $1.31 \leq k_Y \leq 1.37$ the only values reachable with $p \leq 6$ are $4/3$ ($p = 6$) and $101/75$ ($p = 5$). *4/3 is the least exotic candidate*: it needs only sixths, the unit already present in quark hypercharges.

Embeddings. For $U(1)_Y \subset SU(N)$ with fundamental $(3, 1)_a + (1, 2)_b + \text{singlets}$, $k_Y = 2 \text{tr } Y^2$. $SU(5)$ is rigid ($5/3$). $SU(6)$ is not: requiring only that t^c sit in the adjoint leaves $k_Y(a) = 24a^2 - 8a + 4/3$. But $Y = \text{diag}(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3})$, the generator used in the seven-dimensional orbifold model of [12], is the point of *enhanced symmetry* of that family: it commutes with $SU(3) \times SU(3) \subset SU(6)$. It gives $\sin^2 \theta_W(M_U) = 1/(1 + k_Y) = 3/7$ instead of $3/8$. Enumerating hypercharge generators with entries in $\frac{1}{12}\mathbb{Z}$ and at least one Standard Model field in the adjoint yields 23 distinct k_Y in $SU(6)$ and 165 in $SU(7)$, of which exactly one, $4/3$, falls in the window.

A cross-check between frameworks. In that $SU(6)$ model the fundamental contains a doublet $(1, 2)_{-1/3}$ with electric charges $1/6$ and $-5/6$: precisely the unit demanded by the Schellekens condition for $k_Y = 4/3$, although the two frameworks (field theory on an orbifold; world-sheet conformal field theory) are independent. Physically, the heavy $(3, 2)_{2/3}$ gauge bosons carry charges $7/6$ and $1/6$ and cannot decay into Standard Model particles: no gauge-mediated proton decay, but stable fractionally charged relics near M_U .

Two negative results. (a) The weakly coupled heterotic string is the most rigid architecture, since $M_{\text{string}} = 5.27 \times 10^{17} g_{\text{string}} \text{ GeV}$ [18] leaves a single continuous parameter. Fitting it to α_3 at two loops gives $g_{\text{string}} = 0.508$, $M_{\text{string}} = 2.7 \times 10^{17} \text{ GeV}$ and $\alpha^{-1}(0) = 143.8$ ($+4.9\%$) for $k_Y = 4/3$, 160.0 for $5/3$: the familiar string-scale/unification-scale mismatch, seen from the side of α . (b) Intersecting-brane models have no k_Y but linear relations at the string scale. The relation $\alpha_Y^{-1} = \frac{2}{3}\alpha_3^{-1} + \alpha_2^{-1}$ [19] is satisfied by the Standard Model at $4.7 \times 10^{13} \text{ GeV}$; the variant $\alpha_Y^{-1} = \frac{2}{3}\alpha_3^{-1} + \frac{1}{2}\alpha_2^{-1}$ [20] at $1.4 \times 10^{18} \text{ GeV}$, which with M_s set to the reduced Planck mass would give $\alpha^{-1}(0) = 138.3$. With two relations and four reference scales tried, one 1% agreement has a chance probability of order 40%: not significant.

6 Stress tests

6.1 Kaluza–Klein thresholds of the $SU(6)$ model (script 07)

The model is a 7D supersymmetric $SU(6)$ gauge theory on $M^4 \times T^2/Z_6 \times S^1/Z_2$; in 4D language, $V + \Sigma_{1,2,3}$ in the adjoint, with the Z_6 phases and Z_2 parities of [12]. Two counting rules suffice: on T^2/Z_6 each orbit of six non-zero momenta yields one mode for *every* Z_6 charge, while the zero-momentum mode needs trivial charge; on S^1/Z_2 even fields have $n \geq 0$ and odd fields $n \geq 1$. The script reproduces the zero modes stated in the paper (Standard Model gauge bosons, a $U(1)'$ and t^c), which validates the counting, and gives per level

Tower	(b_Y, b_2, b_3)	nature
(i) $p = 0, n \geq 1$	$(8/3, -4, -5)$	massive vectors and one $(3, 1)_{2/3}$ hypermultiplet
(ii) $p \neq 0, n = 0$	$(-16/3, -8, -8)$	$N=2$ multiplets of $SU(5) \times U(1)'$: the 6D fixed planes
(iii) $p \neq 0, n \geq 1$	$(0, 0, 0)$	complete $N=4$ multiplets

Tower (ii) respects $SU(5) \times U(1)'$, not $SU(6)$. It is universal for $SU(2)$ and $SU(3)$ ($b = -8$, the value expected for $N=2$ $SU(5)$ with two fundamental hypermultiplets) but not for hypercharge, which mixes $SU(5)$ with $U(1)'$: $b_Y/k_Y = -4 \neq -8$. Its number of levels grows like $(\Lambda R)^2$. *The relation $k_Y = 4/3$ thus receives a non-universal, power-law correction* and is protected only if the cutoff is close to the compactification scale. An observer extrapolating the Standard Model to the crossing $\alpha_2 = \alpha_3$ would measure

$$k_Y^{\text{apparent}} - \frac{4}{3} \simeq 0.018 S_1 + 0.018 S_2, \quad S_1 = \sum_{n < \Lambda R'} \ln \frac{\Lambda R'}{n}, \quad S_2 = \sum_{\text{orbits, } |p| < \Lambda} \ln \frac{\Lambda}{|p|}. \quad (6)$$

The sign is the right one, and the required $+0.0077 \pm 0.0038$ is obtained for $\Lambda R \simeq 1.2\text{--}1.5$: the residual 0.36 units come out without tuning. But at 2σ one needs $\Lambda R \leq 2.25$ and $\Lambda R' \leq 2.15$, whereas naive dimensional analysis puts 7D strong coupling near $\Lambda R \simeq 17$.

Literature check. That loops on orbifolds generate gauge kinetic terms localised on fixed loci, with coefficients set by the reduced local symmetry, is established [21, 22, 23, 24]. In all cases treated there the fixed loci are points, the localised terms diverge logarithmically, and the power-like corrections are universal bulk effects which, as noted in [24], leave unification untouched. The 7D construction differs because S^1/Z_2 creates codimension-one fixed planes, on which a localised kinetic term diverges quadratically; tower (ii) is exactly that. We found no threshold calculation for this model (its authors note that 7D orbifold breakings had not been studied before). Related string-based bounds on the ratio of cutoff to compactification scale exist [25]. Status: a simple corollary of established principles, apparently not written down for this geometry, not contradicted, *not verified by a specialist*, and computed with a supersymmetric Kaluza–Klein spectrum (Scherk–Schwarz splittings ignored).

Relics. The lightest fractionally charged states have $M \simeq 1/R \simeq 2 \times 10^{16}$ GeV. Thermal production stays below the dark-matter density for $T_{\text{max}} < M/28$ and below 10^{-10} of it for $T_{\text{max}} < M/40 \simeq 5 \times 10^{14}$ GeV, compatible with current bounds on the scale of inflation. No cosmological obstacle, and no expected signal.

6.2 Bayesian evidence with an explicit Occam penalty (script 09)

We computed $Z = \int \pi(\theta) L(\theta) d\theta$ in the Laplace approximation at two loops, with a Gaussian theoretical error σ per coupling and flat priors ($\ln M \in [1 \text{ TeV}, M_{\text{Pl}}]$, $\alpha_U^{-1} \in [0, 100]$, $k_Y \in [1, 2]$ when free).

“Standard Model + 4/3” and “MSSM at 3 TeV + 5/3” tie. Making k_Y continuous costs a factor of seven, which puts a number on the value of a discrete datum. But the fit alone does *not* select

Table 4: Evidence of the architectures for $\sigma = 0.5$ (script 09). The ranking is unchanged for $\sigma = 1$.

Architecture	$\chi^2_{\min}$	$\ln Z$	Bayes factor
MSSM at 3 TeV, $k_Y = 5/3$	0.00	-9.53	1
SM, $k_Y = 4/3$	0.02	-9.63	0.91
SM, $k_Y = 13/10$	0.57	-9.89	0.70
SM, $k_Y = 32/25$	1.28	-10.24	0.49
MSSM, M_S free, $k_Y = 5/3$	0.00	-10.70	0.31
SM, k_Y free (continuous)	0.00	-11.60	0.13
rigid strong coupling (family of Table 2, one loop)	–	-13.5	0.02
SM, $k_Y = 5/3$	27.7	-23.6	8×10^{-7}
weakly coupled heterotic, $k_Y = 4/3$	49.1	-30.9	5×10^{-10}

4/3 over 13/10 or 32/25 once unknown thresholds are allowed for: what singles out 4/3 is the structural argument above, not the statistics.

6.3 Degeneracy with the supersymmetry scale

If the MSSM replaces the Standard Model above M_S , the required k_Y moves continuously from 1.688 ($M_S = 1$ TeV) through 1.457 (10^{10} GeV) to 1.341 (no supersymmetry), see Fig. 1. The two known solutions, 5/3 with supersymmetry at a few TeV and 4/3 with a desert, are the two ends of one line. This is the conservation lemma again: the discrete (k_Y) is exchanged for the continuous (M_S). The conclusion “ $k_Y = 4/3$ ” holds *under the desert hypothesis*.

6.4 The Higgs quartic coupling as an independent observable

With approximate NLO running, $\lambda(M_U) = -0.0152$ for $M_t = 173.34$ GeV (-0.0112 for 172.70, -0.0028 for 171.34), and λ vanishes near $4 \times 10^9 - 2 \times 10^{10}$ GeV; requiring $\lambda(M_U) \geq 0$ would need $M_h \gtrsim 130$ GeV, in agreement with the stability bound of [26, 13], which serves as a check of the code. In the $SU(6)$ model supersymmetry is broken at the compactification scale, which imposes at tree level $0 \leq \lambda(M_U) \leq (g_2^2 + g_Y^2)/8 = 0.060$. The extrapolated value lies outside this window. A loop-sized negative threshold $\delta\lambda \simeq -0.01$ would cure it, so this is a tension and not an exclusion; and placing supersymmetry where λ vanishes gives $k_Y \simeq 1.46$, no longer 4/3.

7 Scorecard: what was recovered, and how

Items that, to our knowledge, are not in the literature in this form, and which should be treated with corresponding caution: the above-the-floor argument against Wyler; the decoy-based statistics of volume formulae; the conservation lemma as stated; the match between the generalised Schellekens unit 1/6 and the $SU(6)$ orbifold exotics; and the non-universal power-law threshold of Section 6.1.

8 What the dialogue did well, and what it got wrong

Errors and retractions (in chronological order).

1. The first system presented $\alpha = Z_0/2R_K$ as relocating the mystery into the vacuum impedance. The second system showed it to be a tautology. Conceded.
2. “ $\alpha(0)$ depends on all charged masses” was loosely stated; corrected to the input/relation distinction of Section 2.
3. The first system quoted a Landau pole near 10^{34} GeV, then 10^{40} , then computed 10^{41} GeV for hypercharge; the factor 1.7 was unaffected, but the drift was flagged to the user when it was noticed.

Table 5: Established results recovered during the session.

Known result	Source	Route taken here
Running of α , infrared floor	textbook	recalled; used as a constraint (Wyler above the floor)
Wyler’s formula; “numerology” critique	[2, 3]	recovered as the unique 10^{-6} hit of a 5.8-million class; critique <i>quantified</i> with decoys
Induced QED gives the right order of magnitude	[6, 7]	redone with hypercharge above M_Z : 82 vs. 137; rephrased as a spectral budget
Strong unification / universal Landau pole	[8, 9, 10]	emerged from a blind scan of 151,263 spectra ($n_5 + 3n_{10} = 6-7$)
SM unifies for $k_Y = 4/3$ at two loops	[11, 12]	independent inputs [13]; expressed as a prediction $\alpha^{-1}(0) = 136.7$
MSSM unification prefers M_S of a few TeV	standard	appears as $k_Y^{\text{req}} = 5/3$ at $M_S \simeq 3$ TeV on a continuous curve
$k_Y < 5/3$ implies fractional charges	[15, 16]	generalised condition coded; $p = 6$ for $4/3$; matched to the $SU(6)$ exotics
String-scale vs. unification-scale mismatch	[18, 17]	one-parameter two-loop fit: $\alpha^{-1}(0) = 143.8$
Brane relation among couplings	[19]	tested on the SM; chance probability estimated
Localised kinetic terms on orbifolds	[21, 22, 23, 24]	extended by mode counting to codimension-one fixed planes
Vacuum stability bound $M_h \gtrsim 129-130$ GeV	[26, 13]	reproduced (130.0) with a simplified NLO code

4. The near-alignment (8.84, 7.88, 8.35) noted by the second system was shown by the first to disappear in $SU(5)$ normalisation, (5.31, 7.88, 8.35), and to be incompatible with complete multiplets; later it was traced to the known near-unification with $k_Y \simeq 1.3$.
5. A one-loop preference for $k_Y = 32/25$ was overturned by the two-loop calculation.
6. A $p \simeq 1-5\%$ “signal” at the Planck scale was reported, explained, and then dissolved by the two-scale scan. In the first run the decoys had been evaluated on a coarser grid than the real world, which biased the comparison; this was noticed and rerun before any conclusion was drawn.
7. A proposed example of a hidden sector generating the hierarchy ($SU(3)$ with 16 flavours) was discarded by the first system itself before being shown, because that theory sits in the conformal window.

What worked. (a) *Decoys everywhere*: every coincidence was compared with what fictitious targets achieve. (b) *A price for every continuous parameter*. (c) *Refusing to scan* when the theoretical error exceeds the spacing of candidates. (d) *Validation by rediscovery*: a search that finds Wyler, or strong unification, without being told, earns some trust for its null results. (e) *Cross-criticism*: the second system’s review produced the best single argument of the study and removed two misleading statements; the first system’s check of that review caught a normalisation artefact and a hidden parameter. Neither system alone would have produced the same document.

What did not. The dialogue reproduced the consensus of the field and did not go beyond it. When asked whether it was worth continuing, the first system answered that further exploration was not: the remaining steps (heterotic M-theory, the quartic threshold with Scherk–Schwarz breaking, a full NNLO Higgs analysis) require specialist machinery in which its errors would no longer be detectable by either participant.

9 Limitations

- One- and two-loop running only; unification-scale thresholds unknown; the conversion of $\alpha_Y(M_t)$ into $\alpha(0)$ is first order.
- The two-loop terms of the top Yukawa and Higgs quartic equations in script 09 are the leading ones, written from the model’s memory and validated only by the agreement of the resulting stability bound with the literature.
- Decoy p -values depend on the chosen family of representations and multiplicity bounds; the two-scale test used only 16 decoys.
- The Kaluza–Klein analysis assumes a supersymmetric spectrum and a hard cutoff, and ignores tree-level localised terms.
- Bibliographic details marked [†] in the reference list were recalled from memory and not re-verified online; the others were checked through web search during the session.
- The three-hour figure is wall-clock time reported by the human participants, not measured by the AI system.

10 Conclusion

The fine-structure constant is very probably not a direct geometric invariant. It remains plausible that $\alpha(0)$ is the output of a renormalisation-group calculation whose ultraviolet inputs are discrete. The conclusion of this exercise is conditional:

If nothing exists between the electroweak scale and 3×10^{16} GeV, then the measured couplings require $k_Y = 4/3$, a value with an identifiable geometric origin (the $U(1)$ commuting with $SU(3) \times SU(3)$ in $SU(6)$) and a definite physical price (superheavy stable states with charges in units of $1/6$). In that world the geometric part of “why 137” is “why that embedding”; the unified coupling $\alpha_U^{-1} \simeq 46$ is a moduli-stabilisation problem, and the last nine units are set by the light spectrum.

Gauge couplings alone cannot distinguish that world from one with supersymmetry at a few TeV. Proton decay (absent for $4/3$ in this construction, expected for $5/3$), superpartner searches and sharper top and Higgs masses can.

As an experiment, the outcome is that a steered dialogue between AI systems with a code sandbox can, within an afternoon, retrace a good part of a thirty-year literature by partly independent routes, attach error bars to its own enthusiasm, and recognise where it should stop. It did not find anything a specialist would call new, and it said so.

Author contributions and disclosure

Fable5.1 wrote and ran all scripts, performed the literature checks and drafted this text, including this sentence. GPT5.6-Sol contributed the critical review integrated in Sections 2–5 (the tautology of $Z_0/2R_K$, the input status of $\alpha(0)$, the above-the-floor argument, the spectral budget and its toy solutions, the three-coupling pole condition, the unification and duality architectures, the precision wall, and the proposal to scan spectra with a parameter penalty). l0d0v1c and pseudoLuc asked the questions, relayed material between the two systems, chose the follow-ups and kept the time. No part of this manuscript has been peer reviewed.

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A Methods

Volumes. Hua’s volumes of the classical domains and of the auxiliary spaces are

$$\begin{aligned} V(\text{I}_{m,n}) &= \pi^{mn} \frac{\prod_{j<m} j! \prod_{j<n} j!}{\prod_{j<m+n} j!}, & V(\text{II}_n) &= \frac{\pi^{n(n+1)/2}}{2^{n(n-1)/2}} \frac{\prod_{k\leq n} (2k)!}{\prod_{k=n}^{2n-1} k!}, \\ V(\text{III}_n) &= \pi^{n(n-1)/2} \frac{\prod_{k\leq n-1} (2k)!}{\prod_{k=n-1}^{2n-3} k!}, & V(\text{IV}_n) &= \frac{\pi^n}{2^{n-1} n!}, \\ V(Q^n) &= \frac{2\pi^{n/2+1}}{\Gamma(n/2)}, & V(S^k) &= \frac{2\pi^{(k+1)/2}}{\Gamma((k+1)/2)}. \end{aligned}$$

Values are handled as logarithms, deduplicated at 10^{-11} and searched by bisection. Domain normalisations are conventional (e.g. $\text{I}(2,2)$ and $\text{IV}(4)$ describe the same space and differ by a factor 16), which is itself an argument against a direct physical reading.

Spectra. One-loop contributions per vector-like Dirac fermion, convention $Q = T_3 + Y$: $E(\frac{4}{3}, 0, 0)$, $L(\frac{2}{3}, \frac{2}{3}, 0)$, $D(\frac{4}{9}, 0, \frac{2}{3})$, $U(\frac{16}{9}, 0, \frac{2}{3})$, $Q(\frac{2}{9}, 2, \frac{4}{3})$, real $T(0, \frac{4}{3}, 0)$, real $G(0, 0, 2)$; supersymmetric version $\times \frac{3}{2}$. $b^{\text{SM}} = (\frac{41}{6}, -\frac{19}{6}, -7)$, $b^{\text{MSSM}} = (11, 1, -3)$. Feasibility of $0 \leq \alpha_i^{-1}(M_Z) - [b_i \ln(\Lambda/M_Z) + \Delta b_i \ln(\Lambda/M)]/2\pi \leq 1$ is decided by scanning $\ln \Lambda$ and intersecting intervals in $\ln(\Lambda/M)$. Inputs at M_Z : $\alpha^{-1} = 127.951$, $\sin^2 \theta_W = 0.23122$, $\alpha_s = 0.1180$.

Two-loop running. Runge–Kutta integration of the gauge equations with $b = (\frac{41}{10}, -\frac{19}{6}, -7)$ and the standard two-loop matrix in $SU(5)$ normalisation, top Yukawa at one loop, from the couplings of [13] at $M_t = 173.34 \text{ GeV}$. MSSM coefficients from [12], $\tan \beta = 10$, no $\overline{\text{MS}}\text{--}\overline{\text{DR}}$ conversion.

Evidence. Laplace approximation, $\ln Z = -\frac{1}{2}\chi_{\min}^2 + \frac{m-n}{2} \ln(2\pi\sigma^2) - \frac{1}{2} \ln \det(J^\top J) - \ln V_{\text{prior}}$ with $n = 3$ observables, m parameters and J the Jacobian at the best fit.

B Python sources

All scripts require only Python 3 and `numpy`. Long lines are wrapped for display (marked $\hookrightarrow$); the code is otherwise reproduced verbatim from the files that produced the numbers in this paper. Indicative run times on a single core: scripts 01, 04, 07: seconds; 02, 05: about a minute; 06, 09: two to three minutes; 03 and 08: several minutes, proportional to the number of decoy worlds requested on the command line.

B.1 01_wyler.py

Wyler’s formula in dimension n and brute-force search (Section 3.3).

```

1  #!/usr/bin/env python3
2  """
3  01 - Wyler's formula: generalisation to dimension n, and brute-force search
4      in the class 2^a 3^b 5^c pi^d (quarter-integer exponents).
5  """
6  import numpy as np, math
7  from math import pi, gamma, factorial
8
9  T = 137.035999177                                # 1/alpha, CODATA 2022
10
11 def VD(n): return pi**n/(2**(n-1)*factorial(n))      # Cartan domain of type IV (Hua)
12 def VQ(n): return 2*pi**(n/2+1)/gamma(n/2)          # Shilov boundary Q^n
13 def VS(k): return 2*pi**((k+1)/2)/gamma((k+1)/2)    # sphere S^k
14
15 print("== Wyler in dimension n:  alpha = 3 V(S^{n-1}) / V(Q^n)^2 * V(D^n)^(1/4) ==")
16 for n in range(3, 9):
17     a = VS(n-1)/VQ(n)*(3/VQ(n))*VD(n)**0.25
18     print(f"  n = {n}      1/alpha = {1/a:.5f}")

```

```

19 print("  closed form n=5:", 2**4.75*3**-1.75*5**-.25*pi**2.75)
20
21 print("\n== Brute-force search 2^a 3^b 5^c pi^d, exponents multiples of 1/4 ==")
22 L = [math.log(x) for x in (2, 3, 5, pi)]; lt = math.log(T)
23 for R in (8, 16, 24):
24     e = np.arange(-R, R+1)/4
25     s01 = (e[:,None]*L[0] + e[None,:]*L[1]).ravel()
26     s23 = np.sort((e[:,None]*L[2] + e[None,:]*L[3]).ravel())
27     for tol in (1e-6, 1e-7, 1e-8):
28         n = (np.searchsorted(s23, lt-s01+tol) - np.searchsorted(s23, lt-s01-tol)).sum()
29         print(f" |exponent| <= {R/4:.0f}    tolerance {tol:.0e} : {int(n)} formula(e)")

```

B.2 02_cartan_volumes.py

Cartan-volume coincidences with decoy targets (Section 3.3).

```

1  #!/usr/bin/env python3
2  """
3  02 - Search for coincidences between physical constants and products of volumes
4        of symmetric spaces (Cartan domains I-IV, spheres, Shilov boundaries),
5        with a look-elsewhere control based on decoy targets.
6  """
7  import numpy as np, itertools
8  from math import pi, factorial as f, gamma, log
9
10 def prod(it):
11     r = 1.0
12     for x in it: r *= x
13     return r
14
15 B = {}
16 for n in range(1, 9):
17     B[f"IV{n}"] = pi**n/(2**(n-1)*f(n))
18 for m, n in [(1,2),(1,3),(1,4),(1,5),(2,2),(2,3),(2,4),(3,3)]:
19     B[f"I{m}{n}"] = prod(f(k) for k in range(1,m))*prod(f(k) for k in range(1,n)) \
20         / prod(f(k) for k in range(1,m+n)) * pi**(m*n)
21 for n in (2, 3):
22     B[f"II{n}"] = pi**(n*(n+1)/2)/2**(n*(n-1)/2) \
23         * prod(f(2*k) for k in range(1,n))/prod(f(k) for k in range(n,2*n))
24 for n in (3, 4, 5):
25     B[f"III{n}"] = pi**(n*(n-1)/2) \
26         * prod(f(2*k) for k in range(1,n-1))/prod(f(k) for k in range(n-1,2*n-2))
27 for k in range(1, 9): B[f"S{k}"] = 2*pi**((k+1)/2)/gamma((k+1)/2)
28 for n in range(2, 9): B[f"Q{n}"] = 2*pi**(n/2+1)/gamma(n/2)
29
30 names = list(B); L = np.array([log(B[k]) for k in names]); nb = len(names)
31 EXP = np.array([-2, -1, -.5, -.25, .25, .5, 1, 2])
32 C = {"1":1, "2":2, "3":3, "4":4, "1/2":.5, "1/3":1/3, "1/4":.25, "3/2":1.5, "2/3":2/3}
33 cn = list(C); lc = np.array([log(v) for v in C.values()])
34 TARGETS = {"1/alpha":137.035999177, "mp/me":1836.152673426,
35            "mmu/me":206.7682827, "mn/me":1838.68366200}
36 lt = {k: log(v) for k, v in TARGETS.items()}
37
38 vals, hits = [], {k: [] for k in TARGETS}
39 for r in (1, 2, 3):
40     for comb in itertools.combinations(range(nb), r):
41         v = lc.copy()
42         for i in comb: v = v[... , None] + L[i]*EXP
43         for k in TARGETS:
44             for ix in np.argwhere(np.abs(v-lt[k]) < 1e-6):
45                 hits[k].append((float(abs(v[tuple(ix)]-lt[k])), comb, tuple(ix)))
46         vals.append(v.ravel())
47 V = np.unique(np.round(np.concatenate(vals), 11))
48 print(nb, "building blocks;", len(V), "distinct values")
49
50 rng = np.random.default_rng(1)
51 for k in TARGETS:
52     print("\n==", k)
53     for tol in (1e-5, 1e-6, 1e-7, 1e-8):
54         n = np.searchsorted(V, lt[k]+tol) - np.searchsorted(V, lt[k]-tol)
55         d = lt[k] + rng.uniform(-.2, .2, 4000)
56         nd = np.searchsorted(V, d+tol) - np.searchsorted(V, d-tol)

```

```

57     print(f"   tol {tol:.0e}: real {n:3d} | decoys, mean {nd.mean():6.2f} | P(decoy >= real) = {np.mean
    ↪ (nd>n):.2f}")
58 for e, comb, ix in sorted(hits[k])[:4]:
59     s = cn[ix[0]] + " * " + " * ".join(f"{names[i]}^{EXP[j]:g}" for i, j in zip(comb, ix[1:]))
60     print(f"       deviation {e:.1e}: {s}")

```

B.3 03_spectrum_scan.py

Scan of vector-like spectra with decoy worlds (Section 4.2).

```

1  #!/usr/bin/env python3
2  """
3  03 - Scan of vector-like spectra: which matter contents make the three gauge couplings
4  simultaneously strong (0 <= alpha_i^-1(Lambda) <= 1) at a common scale Lambda?
5  One loop, a single common mass M >= 1 TeV. Control: decoy worlds.
6
7  Usage: python3 03_spectrum_scan.py [number_of_decoys]   (default 20; 80 were used for the paper,
8  allow several minutes per configuration)
9  """
10 import numpy as np, itertools, math, sys
11 tp = 2*math.pi; MZ = 91.1876
12 AINV, S2W, AS = 127.951, 0.23122, 0.1180
13 A = np.array([AINV*(1-S2W), AINV*S2W, 1/AS])          # 1/alpha_Y, 1/alpha_2, 1/alpha_3 at M_Z
14 bSM = np.array([41/6, -19/6, -7.0]); bMSSM = np.array([11.0, 1.0, -3.0]) # convention Q = T3 + Y
15 # (db_Y, db_2, db_3) per vector-like Dirac fermion; T and G: real fermions
16 FIELDS = {"E":(4/3,0,0), "L":(2/3,2/3,0), "D":(4/9,0,2/3), "U":(16/9,0,2/3),
17           "Q":(2/9,2,4/3), "T":(0,4/3,0), "G":(0,0,2)}
18 names = list(FIELDS); C = np.array(list(FIELDS.values()))
19 S = np.array(list(itertools.product(*[range(7)]*5, range(3), range(3))))
20 comp = S.sum(1)
21 su5 = (S[:,1]==S[:,2]) & (S[:,0]==S[:,3]) & (S[:,0]==S[:,4]) & (S[:,5]==0) & (S[:,6]==0)
22
23 def feasible(A, susy, xlo, xhi, nx=40, Mmin=1000.):
24     """For each spectrum: is there (Lambda, M) such that alpha_i^-1(Lambda) lies in [0,1] for all i?
25     x = ln(Lambda/MZ), y = ln(Lambda/M). At fixed x, each coupling imposes an interval in y."""
26     db = S@C*(1.5 if susy else 1.0)          # supermultiplets: x 3/2
27     if susy: db = db + (bMSSM - bSM)        # superpartners at the same scale M
28     x = np.linspace(xlo, xhi, nx); ymax = x - math.log(Mmin/MZ)
29     lo = np.zeros((len(S), nx)); hi = np.broadcast_to(ymax, (len(S), nx)).copy()
30     ok = np.ones((len(S), nx), bool)
31     for i in range(3):
32         up = tp*A[i] - bSM[i]*x; dn = up - tp          # db_i * y must fall in [dn, up]
33         d = db[:, i][:, None]
34         with np.errstate(divide='ignore', invalid='ignore'):
35             l = np.where(d > 0, dn/d, -np.inf); h = np.where(d > 0, up/d, np.inf)
36             ok &= np.where(d > 0, True, (dn <= 0) & (up >= 0))
37             lo = np.maximum(lo, l); hi = np.minimum(hi, h)
38     f = ok & (lo <= hi); any_ = f.any(1)
39     ix = np.where(any_, (f*np.arange(nx)).sum(1)//np.maximum(f.sum(1), 1), 0)
40     r = np.arange(len(S))
41     return any_, x[ix], (lo[r, ix] + hi[r, ix])/2
42
43 if __name__ == "__main__":
44     ndec = int(sys.argv[1]) if len(sys.argv) > 1 else 20
45     X = lambda lam: math.log(lam/MZ)
46     windows = {"Lambda free 1e10 - 1.22e19 GeV": (X(1e10), X(1.22e19), 60),
47               "Planck window 2.4e18 - 1.22e19 GeV": (X(2.435e18), X(1.22e19), 30)}
48     rng = np.random.default_rng(11)
49     print(len(S), "spectra; couplings at M_Z:", np.round(A, 3))
50     for susy in (False, True):
51         for wn, (a, b, nx) in windows.items():
52             ok, xb, yb = feasible(A, susy, a, b, nx)
53             print(f"\n### base {'MSSM (superpartners at M)' if susy else 'Standard Model'} | {wn}")
54             print(f"   compatible spectra: {ok.sum()} ({100*ok.mean():.2f} %); minimal complexity: {comp[ok
    ↪ ].min()}")
55             for w in (1.25, 1.5, 2.0):
56                 dec = A*np.exp(rng.uniform(-math.log(w), math.log(w), (ndec, 3)))
57                 nd = np.array([feasible(d, susy, a, b, nx)[0].sum() for d in dec])
58                 print(f"   decoys x{w}<4: median {np.median(nd):6.0f} | P(decoy >= real) = {np.mean(nd >=
    ↪ ok.sum()):.3f}")
59             for j in np.where(ok)[0][np.argsort(comp[ok])[:5]:
60                 lam = MZ*math.exp(xb[j]); M = lam*math.exp(-yb[j])

```

```

61         print("      ", {n: int(v) for n, v in zip(names, S[j]) if v}, f"Lambda ~ {lam:.1e} M ~ {M
    ↪      :.1e} GeV")
62         print("      complete SU(5) multiplets (n5, n10):", [(int(S[j][1]), int(S[j][0])) for j in np.
    ↪      where(ok & su5)[0]])
63         # test of the interpretation: the excess in the Planck window comes from the near-equality of the
    ↪      required db
64         print("\n### 'democracy' test (SM base, Planck window): required R_i = db_i * y")
65         a, b, nx = windows["Planck window 2.4e18 - 1.22e19 GeV"]; x0 = (a+b)/2
66         print("      real world: R =", np.round(tp*A - bSM*x0, 1))
67         for R in [(330,330,330), (420,330,250), (250,330,420), (500,330,160), (160,330,500)]:
68             Ad = (np.array(R) + bSM*x0)/tp
69             print(f"      R = {R} -> {feasible(Ad, False, a, b, nx)[0].sum()} spectra")

```

B.4 04_strong_coupling_prediction.py

Prediction of $1/\alpha(0)$ under common strong coupling (Section 4.3).

```

1  #!/usr/bin/env python3
2  """
3  04 - 'Common strong coupling' scenario: MSSM + n complete SU(5) multiplets (n = n5 + 3 n10)
4  at the scale M (superpartners included),  $\alpha_i^{-1}(\text{Lambda}) = c * k_i$  with c in [0,1].
5  Lambda is fitted to alpha_s; alpha_2, alpha_Y,  $\sin^2(\theta_W)$  and  $1/\alpha(0)$  are predicted.
6  Shows the hypersensitivity to M:  $d(1/\alpha(0))/d \ln M \sim 11$ . One loop.
7  """
8  import math
9  tp = 2*math.pi; MZ = 91.1876
10 A = dict(Y=98.366, w=29.585, s=8.4746) # 1/alpha_i(M_Z)
11 D_IR = 137.035999 - 127.951 # running from M_Z down to 0
12 bSM = dict(Y=41/6, w=-19/6, s=-7.0); bMSSM = dict(Y=11.0, w=1.0, s=-3.0)
13 k = dict(Y=5/3, w=1.0, s=1.0) # one complete multiplet: db_i = k_i
14
15 print(" n   M(TeV)   c   Lambda(GeV)   1/a_2   1/a_Y   sin2_W   1/alpha(0)")
16 for n in (5, 6, 7):
17     for M in (1e3, 3e3, 1e4, 3e4):
18         for c in (0, 0.5, 1):
19             t = math.log(M/MZ)
20             Bh = {i: bMSSM[i] + n*k[i] for i in A}
21             xt = (tp*(A['s'] - c) - bSM['s']*t)/Bh['s'] # ln(Lambda/M) fitted to alpha_s
22             p = {i: c*k[i] + (bSM[i]*t + Bh[i]*xt)/tp for i in A}
23             am = p['Y'] + p['w']
24             print(f" {n}   {M/1e3:5.0f}   {c:3.1f}   {M*math.exp(xt):10.2e}   {p['w']:6.2f}   {p['Y']:6.2f}
    ↪   {p['w']/am:.4f}   {am + D_IR:8.2f}")

```

B.5 05_kY_two_loops.py

Two-loop determination of k_Y and prediction of $1/\alpha(0)$ (Section 5.1).

```

1  #!/usr/bin/env python3
2  """
3  05 - Hypercharge normalisation k_Y and prediction of 1/alpha(0),
4  Standard Model only, at one and two loops.
5
6  Principle: impose  $\alpha_Y^{-1} / k_Y = \alpha_2^{-1} = \alpha_3^{-1} = \alpha_U^{-1}$  at M_U.
7  (M_U, alpha_U) are fitted to the measured alpha_2 and alpha_3; alpha_Y is then PREDICTED,
8  hence 1/alpha(0) as well:  $1/\alpha(0)_{\text{pred}} = 137.036 + [\alpha_Y^{-1} \text{pred} - \alpha_Y^{-1} \text{obs}](M_t)$ .
9
10 Inputs: MS-bar couplings at  $\mu = M_t = 173.34$  GeV, extracted at NNLO by
11 Buttazzo et al., JHEP 12 (2013) 089 [arXiv:1307.3536], eqs (57)-(59):
12 g_Y = 0.35830, g_2 = 0.64779, y_t = 0.93690,
13 g_3 = 1.1666 + 0.00314*(alpha_s(MZ)-0.1184)/0.0007
14 Equations: two-loop SM gauge RGEs (Machacek-Vaughn), one-loop top Yukawa.
15 Internal normalisation:  $g_1^2 = (5/3) g_Y^2$  (a computational convention, with no effect on k_Y).
16 """
17 import numpy as np, math
18 from fractions import Fraction as Fr
19
20 MT = 173.34
21 ALPHA0_INV = 137.035999177
22 b = np.array([41/10, -19/6, -7.0])
23 B = np.array([[199/50, 27/10, 44/5],
24               [9/10, 35/6, 12.0],

```

```

25     [11/10, 9/2, -26.0]])
26 Ct = np.array([17/10, 3/2, 2.0])
27 K = 1/(16*math.pi**2)
28
29 def beta(y, loops):
30     g, yt = y[:3], y[3]
31     dg = K*b*g**3
32     if loops == 2:
33         dg = dg + K*K*g**3*(B0(g**2) - Ct*yt**2)
34     dyt = K*yt*(4.5*yt**2 - 17/20*g[0]**2 - 9/4*g[1]**2 - 8*g[2]**2)
35     return np.append(dg, dyt)
36
37 def run(y0, t0, t1, loops, n=4000):
38     """Runge-Kutta 4 from t0 to t1, t = ln(mu/GeV)."""
39     y, h = np.array(y0, float), (t1-t0)/n
40     for _ in range(n):
41         k1 = beta(y, loops); k2 = beta(y+h/2*k1, loops)
42         k3 = beta(y+h/2*k2, loops); k4 = beta(y+h*k3, loops)
43         y = y + h/6*(k1+2*k2+2*k3+k4)
44     return y
45
46 ainv = lambda g: 4*math.pi/g**2
47
48 def inputs(alpha_s=0.1180, dyt=0.0):
49     gY, g2, yt = 0.35830, 0.64779, 0.93690+dyt
50     g3 = 1.1666 + 0.00314*(alpha_s-0.1184)/0.0007
51     return np.array([math.sqrt(5/3)*gY, g2, g3, yt])
52
53 def k_requis(y0, loops):
54     """Run the couplings up, find M_U where alpha_2 = alpha_3, return (M_U, alpha_U^-1, required k_Y)."""
55     t, y, h = math.log(MT), y0.copy(), 0.05
56     while True:
57         yn = run(y, t, t+h, loops, n=20)
58         if ainv(yn[1]) - ainv(yn[2]) < 0: # crossing inside [t, t+h]: bisection
59             lo, hi = 0.0, h
60             for _ in range(40):
61                 mid = (lo+hi)/2; ym = run(y, t, t+mid, loops, n=20)
62                 if ainv(ym[1]) - ainv(ym[2]) < 0: hi = mid
63                 else: lo = mid
64             ym = run(y, t, t+lo, loops, n=20)
65             aU = ainv(ym[1]); aYinv = (5/3)*ainv(ym[0])
66             return math.exp(t+lo), aU, aYinv/aU, ym[3]
67     t, y = t+h, yn
68
69 def prediction(kY, y0, loops):
70     """Fit (M_U, alpha_U) to alpha_2, alpha_3 at M_t by Newton iteration; predict alpha_Y^-1(M_t)."""
71     MU, aU, _, ytU = k_requis(y0, loops)
72     p = np.array([math.log(MU), aU])
73     target = np.array([ainv(y0[1]), ainv(y0[2])])
74     def down(p):
75         gU = math.sqrt(4*math.pi/p[1])
76         gYU = math.sqrt(4*math.pi/(kY*p[1]))
77         y = run([math.sqrt(5/3)*gYU, gU, gU, ytU], p[0], math.log(MT), loops, n=3000)
78         return y
79     for _ in range(12):
80         y = down(p); r = np.array([ainv(y[1]), ainv(y[2])]) - target
81         if np.abs(r).max() < 1e-10: break
82         J = np.zeros((2,2))
83         for j, e in enumerate((1e-4, 1e-4)):
84             dp = p.copy(); dp[j] += e; yy = down(dp)
85             J[:,j] = (np.array([ainv(yy[1]), ainv(yy[2])]) - target - r)/e
86         p = p - np.linalg.solve(J, r)
87     y = down(p)
88     aY_pred = (5/3)*ainv(y[0]); aY_obs = (5/3)*ainv(y0[0])
89     return math.exp(p[0]), p[1], ALPHAO_INV + aY_pred - aY_obs
90
91 if __name__ == "__main__":
92     y0 = inputs()
93     print("Inputs at M_t: 1/alpha_Y = %.3f 1/alpha_2 = %.3f 1/alpha_3 = %.3f"
94           % ((5/3)*ainv(y0[0]), ainv(y0[1]), ainv(y0[2])))
95     print("\n== k_Y required for exact unification ==")
96     for loops in (1, 2):
97         MU, aU, k, _ = k_requis(y0, loops)
98         print(f"{loops} loop(s): M_U = {MU:.3e} GeV 1/alpha_U = {aU:.3f} k_Y = {k:.4f}")
99     print("\n== uncertainties on k_Y at two loops ==")

```

```

100 k0 = k_requis(y0, 2)[2]
101 for lab, yy in [("alpha_s +0.0009", inputs(0.1189)), ("alpha_s -0.0009", inputs(0.1171)),
102               ("y_t +0.005 (~ M_t +1 GeV)", inputs(dyt=0.005))]:
103     print(f" {lab:28s} dk_Y = {k_requis(yy,2)[2]-k0:+.4f}")
104 print("\n== prediction of 1/alpha(0) as a function of k_Y ==")
105 print(" k_Y      1 loop      2 loops      dev. 2l      M_U (2l)")
106 for fr in (Fr(5,3), Fr(4,3), Fr(13,10), Fr(32,25), Fr(5,4), Fr(6,5), Fr(1,1)):
107     r1 = prediction(float(fr), y0, 1); r2 = prediction(float(fr), y0, 2)
108     print(f" {str(fr):6s} {r1[2]:9.2f} {r2[2]:9.2f} {100*(r2[2]/ALPHA0_INV-1):+6.2f} % {r2
    ↪ [0]:.2e}")

```

B.6 06_kY_origin.py

Origin of $k_Y = 4/3$: Schellekens condition, embeddings, heterotic and brane tests (Section 5.2).

```

1  #!/usr/bin/env python3
2  """
3  06 - Where can  $k_Y = 4/3$  come from? Four tests.
4
5  A. Generalised Schellekens condition (heterotic strings, levels  $k_2 = k_3 = 1$ ):
6  electric charges quantised in units of  $1/p$  are possible iff
7   $[p \bmod 3 \neq 0]/3 + [p \bmod 2 \neq 0]/4 + p^2 k_Y/4$  is an integer.
8  (Dienes, Faraggi, March-Russell, hep-th/9510223, eqs 6.30-6.31;  $p = 1$  gives back
9   $k_3/3 + k_2/4 + k_Y/4 = 0 \bmod 1$ , i.e.  $k_Y \geq 5/3$ .)
10 B. Embeddings of  $U(1)_Y$  in  $SU(N)$ :  $k_Y = 2 \operatorname{tr}(Y^2)$  on the fundamental
11  $N = (3,1)_a + (1,2)_b + \text{singlets } s_j$ , traceless. Rigid or not?
12 C. Weakly coupled heterotic string:  $M_{\text{string}} = 5.27e17 * g_{\text{string}}$  GeV (Kaplanovsky),
13 a single continuous parameter. Two-loop predictions.
14 D. Brane relations (type II) at the string scale:
15 (A)  $1/\alpha_Y = (2/3)/\alpha_3 + 1/\alpha_2$  (Blumenhagen-Lust-Stieberger 2003)
16 (B)  $1/\alpha_Y = (2/3)/\alpha_3 + (1/2)/\alpha_2$  (Pati-Salam-like variant)
17 At which scale does the Standard Model satisfy them?
18 """
19 import numpy as np, math, itertools
20 from fractions import Fraction as Fr
21 from math import gcd
22
23 # ----- RGE core (identical to script 05) -----
24 MT = 173.34; ALPHA0_INV = 137.035999177
25 b = np.array([41/10, -19/6, -7.0])
26 B = np.array([[199/50, 27/10, 44/5], [9/10, 35/6, 12.0], [11/10, 9/2, -26.0]])
27 Ct = np.array([17/10, 3/2, 2.0]); K = 1/(16*math.pi**2)
28 def beta(y):
29     g, yt = y[:3], y[3]
30     dg = K*b*g**3 + K*K*g**3*(B0(g**2) - Ct*yt**2)
31     return np.append(dg, K*yt*(4.5*yt**2 - 17/20*g[0]**2 - 9/4*g[1]**2 - 8*g[2]**2))
32 def run(y0, t0, t1, n=3000):
33     y, h = np.array(y0, float), (t1-t0)/n
34     for _ in range(n):
35         k1 = beta(y); k2 = beta(y+h/2*k1); k3 = beta(y+h/2*k2); k4 = beta(y+h*k3)
36         y = y + h/6*(k1+2*k2+2*k3+k4)
37     return y
38 ainv = lambda g: 4*math.pi/g**2
39 g3 = 1.1666 + 0.00314*(0.1180-0.1184)/0.0007
40 Y0 = np.array([math.sqrt(5/3)*0.35830, 0.64779, g3, 0.93690]) # couplings at M_t
41 obs = np.array([(5/3)*ainv(Y0[0]), ainv(Y0[1]), ainv(Y0[2])]) # 1/alpha_Y, 1/alpha_2, 1/alpha_3
42
43 # ----- A. generalised Schellekens condition -----
44 def ok(p, k): return ((Fr(1,3) if p % 3 else 0) + (Fr(1,4) if p % 2 else 0) + Fr(p*p)*k/4).denominator ==
    ↪ 1
45 def pmin(k, pmax=200):
46     return next((p for p in range(1, pmax+1) if ok(p, k)), None)
47 print("== A. Minimal charge unit 1/p required by  $k_Y$  (heterotic, level 1) ==")
48 for k in (Fr(5,3), Fr(4,3), Fr(13,10), Fr(32,25), Fr(5,4), Fr(7,6), Fr(1)):
49     print(f" k_Y = {str(k):6s} -> p_min = {pmin(k)}")
50 lo, hi = Fr(131,100), Fr(137,100)
51 print(f" k_Y allowed in the window [{float(lo)}, {float(hi)}] (required k at 2 loops: 1.341 +- 0.004 +
    ↪ thresholds):")
52 for P in (6, 12):
53     found = sorted({Fr(n, d) for p in range(1, P+1) for d in (1,2,3,4,5,6,9,12,16,18,25,36,48,75,100,144)
54                    for n in range(d, 2*d+1) if lo <= Fr(n, d) <= hi and pmin(Fr(n, d), P) is not None})
55     print(f" p <= {P:2d} : " + ", ".join(f"{str(k)} (p={pmin(k)})" for k in found))
56

```

```

57 # ----- B. embeddings in SU(N) -----
58 print("\n== B. k_Y = 2 tr(Y^2) for U(1)_Y in SU(N) ==")
59 print("    SU(5) : Y = diag(-1/3,-1/3,-1/3,1/2,1/2), unique up to normalisation -> k_Y =", 2*(3*Fr(1,9)+2*
    ↪ Fr(1,4)))
60 print("    SU(6), with t^c = (3bar,1)_{-2/3} in the adjoint (a - s = 2/3): ONE-parameter family")
61 print("        k_Y(a) = 24 a^2 - 8 a + 4/3; examples:", {str(a): str(24*a*a-8*a+Fr(4,3)) for a in (Fr(0)
    ↪ , Fr(1,6), Fr(1,4), Fr(1,3), Fr(5,12))})
62 SMADJ = {"Q":("32", Fr(1,6)), "u^c":("31", Fr(2,3)), "d^c":("31", Fr(1,3)), "H/L":("21", Fr(1,2)), "e^c":(
    ↪ "11", Fr(1))}
63 for N in (6, 7):
64     ns = N-5; grid = [Fr(i, 12) for i in range(-12, 13)]
65     ks = {}
66     for a in grid:
67         for b_ in grid:
68             for s in itertools.combinations_with_replacement(grid, ns-1):
69                 last = -(3*a + 2*b_ + sum(s)); ss = tuple(s) + (last,)
70                 if abs(last) > 1: continue
71                 diffs = {"32": {abs(a-b_)}, "31": {abs(a-x) for x in ss}, "21": {abs(b_-x) for x in ss},
72                         "11": {abs(x-y) for x in ss for y in ss}}
73                 if not any(v in diffs[t] for t, v in SMADJ.values()): continue # at least one SM field
    ↪ in the adjoint
74                 k = 2*(3*a*a + 2*b_*b_ + sum(x*x for x in ss))
75                 if k > 0: ks.setdefault(k, (a, b_, ss))
76 win = sorted(k for k in ks if lo <= k <= hi)
77 print(f"    SU({N}), charges multiples of 1/12: {len(ks)} distinct values of k_Y, of which {len(win)}
    ↪ in the window:",
78       ", ".join(str(k) for k in win[:12]), "... if len(win) > 12 else """)
79 print("    Barger et al.: SU(6) diag(1/3,1/3,1/3,-1/3,-1/3,-1/3) -> 4/3; exotic doublet with charges 1/6
    ↪ and -5/6 (p = 6, cf. A)")
80
81 # ----- C. weakly coupled heterotic string -----
82 print("\n== C. Heterotic string: k_Y g_Y^2 = g_2^2 = g_3^2 = g_s^2 at M_s = 5.27e17 g_s GeV (2 loops) ==")
83 ytop = run(Y0, math.log(MT), math.log(3e17))[3]
84 def down(gs, kY):
85     Ms = 5.27e17*gs
86     return run([math.sqrt(5/3)*gs/math.sqrt(kY), gs, gs, ytop], math.log(Ms), math.log(MT))
87 for kY in (Fr(5,3), Fr(4,3)):
88     lo_g, hi_g = 0.45, 0.60
89     for _ in range(50):
90         mid = (lo_g+hi_g)/2
91         if ainv(down(mid, float(kY)))[2] > obs[2]: lo_g = mid
92         else: hi_g = mid
93     y = down(mid, float(kY)); pred = np.array([(5/3)*ainv(y[0]), ainv(y[1])])
94     a0 = ALPHAO_INV + (pred - obs[:2]).sum()
95     print(f"    k_Y = {str(kY):4s} : g_s = {mid:.3f}, M_s = {5.27e17*mid:.2e} GeV | 1/a_2 = {pred[1]:.2f} (
    ↪ obs {obs[1]:.2f}), "
96           f"1/a_Y = {pred[0]:.2f} (obs {obs[0]:.2f}) | 1/alpha(0) = {a0:.1f} ({100*(a0/ALPHAO_INV-1):+.1f}
    ↪ %)")
97
98 # ----- D. brane relations -----
99 print("\n== D. Brane relations: scale at which the SM satisfies them (2 loops) ==")
100 ts = np.linspace(math.log(MT), math.log(1.3e19), 400); ys = [Y0]
101 for t0, t1 in zip(ts[:-1], ts[1:]): ys.append(run(ys[-1], t0, t1, n=10))
102 ys = np.array(ys); aY, a2, a3 = (5/3)*ainv(ys[:,0]), ainv(ys[:,1]), ainv(ys[:,2])
103 REF = {"M_U (a2=a3)":2.9e16, "heterotic M_s":2.7e17, "reduced M_Planck":2.435e18, "M_Planck":1.22e19}
104 for name, f in {"(A) 2/3, 1": aY - (2/3)*a3 - a2, "(B) 2/3, 1/2": aY - (2/3)*a3 - 0.5*a2}.items():
105     i = np.where(np.sign(f[:-1]) != np.sign(f[1:]))[0]
106     Ms = math.exp(np.interp(0, [f[i[0]+1], f[i[0]]], [ts[i[0]+1], ts[i[0]]])) if len(i) else float('nan')
107     print(f"    relation {name:13s} satisfied at M_s = {Ms:.2e} GeV")
108     for rn, M in REF.items():
109         d = np.interp(math.log(M), ts, f) # predicted 1/alpha(0) if M_s were imposed
110         print(f"        if M_s = {rn:17s}: 1/alpha(0) = {ALPHAO_INV - d:7.2f} ({-100*d/ALPHAO_INV:+.1f}
    ↪ %)")

```

B.7 07_KK_thresholds_SU6.py

Kaluza-Klein thresholds of the 7D $SU(6)$ model and relics (Section 6.1).

```

1 #!/usr/bin/env python3
2 """
3 07 - Kaluza-Klein thresholds of the 7D SU(6) model (Barger, Jiang, Langacker, Li,
4     hep-ph/0504093, section III.B) on M^4 x T^2/Z_6 x S^1/Z_2, and an estimate of the
5     relic abundance of fractionally charged states.

```

```

6
7 4D content of the 7D vector multiplet: V + Sigma_1 + Sigma_2 + Sigma_3, all in the adjoint 35.
8 Blocks of the adjoint (1 = colour, 2 = doublet, 3 = singlet), Z_6 phases and Z_2 parities
9 read off eqs (32)-(35) of the paper with n_1 = 2, n_2 = 5.
10
11 Mode-counting rules:
12 * on T^2/Z_6, each orbit of 6 non-zero momenta provides ONE mode for EACH Z_6 charge;
13   the zero-momentum mode exists only for trivial Z_6 charge;
14 * on S^1/Z_2, parity +: n >= 0; parity -: n >= 1.
15 Beta coefficients (convention Q = T_3 + Y, N=1 multiplets): vector -3 T, chiral +T.
16 """
17 import numpy as np, math
18 from fractions import Fraction as Fr
19
20 w = lambda k: k % 6
21 n1, n2 = 2, 5
22 ph = {(1,1):0, (2,2):0, (3,3):0, (1,2):-n1, (2,1):n1, (1,3):-n2, (3,1):n2, (2,3):n1-n2, (3,2):n2-n1}
23 typeA = {(1,1),(2,2),(3,3),(1,2),(2,1)} # Z_2 parity of V: +; other blocks: -
24 fields = {"V":(0,+1), "S1":(-1,+1), "S2":(0,-1), "S3":(+1,-1)} # (phase shift, sign relative to V)
25 # (tr Y^2, T_SU2, T_SU3) of each block, Y = diag(1/3,1/3,1/3,-1/3,-1/3,-1/3)
26 idx = {(1,1):(Fr(0),Fr(0),Fr(3)), (2,2):(Fr(0),Fr(2),Fr(0)), (3,3):(Fr(0),)*3,
27        (1,2):(Fr(8,3),Fr(3,2),Fr(1)), (2,1):(Fr(8,3),Fr(3,2),Fr(1)),
28        (1,3):(Fr(4,3),Fr(0),Fr(1,2)), (3,1):(Fr(4,3),Fr(0),Fr(1,2)),
29        (2,3):(Fr(0),Fr(1,2),Fr(0)), (3,2):(Fr(0),Fr(1,2),Fr(0))}
30 def content(select):
31     b = [Fr(0)]*3; lst = []
32     for f, (dk, sg) in fields.items():
33         for blk, k in ph.items():
34             par = (+1 if blk in typeA else -1)*sg
35             if select(w(k+dk), par):
36                 c = -3 if f == "V" else 1
37                 b = [x + c*y for x, y in zip(b, idx[blk])]; lst.append(f"{f}-{blk}")
38     return b, lst
39 b0, l0 = content(lambda k, par: k == 0 and par == +1) # zero modes
40 bI, lI = content(lambda k, par: k == 0) # tower p = 0, n >= 1
41 bII, lII = content(lambda k, par: par == +1) # towers p != 0, n = 0
42 bIII, _ = content(lambda k, par: True) # p != 0, n >= 1: complete N=4 multiplet
43 print("zero modes : ", l0)
44 print("tower (i) p=0, n>=1 : (b_Y, b_2, b_3) =", [str(x) for x in bI])
45 print("tower (ii) p!=0, n=0 : (b_Y, b_2, b_3) =", [str(x) for x in bII], " -> non-universal: b_Y/k_Y ="
46       "\n", bII[0]*Fr(3,4))
47 print("tower (iii) p!=0, n>=1 : (b_Y, b_2, b_3) =", [str(x) for x in bIII], " (N=4)")
48 # ---- shift of the apparent k_Y ----
49 bSM = np.array([41/6, -19/6, -7.0]); aU = 46.18; kY = 4/3
50 def S1(LRp): return sum(math.log(LRp/n) for n in range(1, int(LRp)+1) if n < LRp)
51 def S2(LRp):
52     s, m = 0.0, int(LRp)+2
53     for a in range(-m, m+1):
54         for c in range(-m, m+1):
55             N = a*a - a*c + c*c
56             if N and (2/math.sqrt(3))*math.sqrt(N) < LRp: s += math.log(LRp/((2/math.sqrt(3))*math.sqrt(N)))
57             ↪ /6
58     return s
59 def dk(LRp, LRp):
60     s1, s2 = S1(LRp), S2(LRp)
61     D = (np.array([float(x) for x in bI])*s1 + np.array([float(x) for x in bII])*s2)/(2*math.pi)
62     lnMU = (D[1]-D[2])/((bSM[1]-bSM[2])/(2*math.pi)) # ln(apparent M_U / Lambda)
63     aY = kY*aU + D[0] - bSM[0]*lnMU/(2*math.pi); a2 = aU + D[1] - bSM[1]*lnMU/(2*math.pi)
64     return aY/a2 - kY, lnMU
65 print("\napparent k_Y - 4/3 (required by the two-loop analysis: +0.0077 +- 0.0038)")
66 grid = (1.0, 1.2, 1.5, 2.0, 3.0, 5.0)
67 print(" Lambda*R' \ \ Lambda*R : " + " ".join(f"{x:6.1f}" for x in grid))
68 for LRp in grid:
69     print(f" {LRp:4.1f} : " + " ".join(f"{dk(LRp, LRp)[0]:+6.3f}" for LRp in grid))
70 ok = [(LR, LRp) for LR in np.arange(1.0, 6.01, 0.05) for LRp in np.arange(1.0, 6.01, 0.05)
71       if 0.0001 <= dk(LR, LRp)[0] <= 0.0153]
72 print(f" region compatible at 2 sigma: Lambda*R' <= {max(x for x, _ in ok):.2f}, Lambda*R' <= {max(y for _,
73       ↪ y in ok):.2f}")
74 # 7D strong-coupling bound (naive dimensional analysis)
75 l7 = (4*math.pi)**3.5*math.gamma(3.5); vol = (math.sqrt(3)/2)*(2*math.pi)**2/6*math.pi
76 print(f" 7D strong coupling (NDA): Lambda*R ~ {(17*aU/(4*math.pi)/vol)**(1/3):.0f} (for R = R')")
77 # ---- fractionally charged relics ----
78 print("\nStable fractionally charged relics (mass M ~ 1/R ~ 2e16 GeV), thermal production")

```

```

78 MPl, gs, M, al, Nch, g = 1.22e19, 106.75, 2e16, 1/46.2, 10, 12
79 def omega(x):
80     T = M/x; neq2 = g*g*(M*T/(2*math.pi))**3*math.exp(-2*x)
81     Y = (math.pi*al*al*Nch/M**2)*neq2/(2*x) / ((2*math.pi**2/45)*gs*T**3 * 1.66*math.sqrt(gs)*T**2/MPl)
82     return M*Y/3.6e-9
83 for target, lab in ((0.12, "all of the dark matter"), (1.2e-11, "1e-10 of the dark matter")):
84     lo, hi = 1.0, 200.0
85     for _ in range(60):
86         mid = (lo+hi)/2
87         if omega(mid) > target: lo = mid
88         else: hi = mid
89     print(f" Omega h^2 < {target:.1e} ({lab}): T_max < M/{mid:.0f} = {M/mid:.1e} GeV")

```

B.8 08_two_scale_scan.py

Two-scale extension of the spectrum scan (Section 4.2).

```

1  #!/usr/bin/env python3
2  """
3  08 - Extension of scan 03 to TWO mass scales: M_c for coloured multiplets (D, U, Q, G),
4       M_u for uncoloured ones (E, L, T). Standard Model base, one loop.
5       Same condition: 0 <= alpha_i^-1(Lambda) <= 1 for i = Y, 2, 3, with M_c, M_u >= 1 TeV.
6       Question: does the mild preference for our world in the Planck window (script 03) survive
7       the addition of one continuous parameter?
8
9  Usage: python3 08_two_scale_scan.py [number_of_decoys] (default 20; ~15 s per evaluation)
10 """
11 import numpy as np, itertools, math, sys, time
12 tp = 2*math.pi; MZ = 91.1876
13 A = np.array([127.951*(1-0.23122), 127.951*0.23122, 1/0.1180])
14 bSM = np.array([41/6, -19/6, -7.0])
15 FIELDS = {"E":(4/3,0,0), "L":(2/3,2/3,0), "D":(4/9,0,2/3), "U":(16/9,0,2/3), "Q":(2/9,2,4/3), "T"
16           ↪ : (0,4/3,0), "G":(0,0,2)}
17 names = list(FIELDS); C = np.array(list(FIELDS.values()), dtype=np.float32)
18 S = np.array(list(itertools.product(*[range(7)]*5, range(3), range(3))), dtype=np.float32)
19 col = np.array([n in "DUQG" for n in names])
20 DC = S[:, col] @ C[col]; DU = S[:, ~col] @ C[~col] # coloured / uncoloured contributions
21
22 def feasible2(A, xlo, xhi, nx, ny=60, chunk=6000, Mmin=1000.):
23     x = np.linspace(xlo, xhi, nx, dtype=np.float32); ymax = x - np.float32(math.log(Mmin/MZ))
24     y2 = np.linspace(0, 1, ny, dtype=np.float32)[None, :]*ymax[:, None] # (nx, ny)
25     out = np.zeros(len(S), bool)
26     for s in range(0, len(S), chunk):
27         dc, du = DC[s:s+chunk], DU[s:s+chunk]
28         lo = np.zeros((len(dc), nx, ny), np.float32); hi = np.broadcast_to(ymax[None, :, None], lo.shape).
29         ↪ copy()
30         ok = np.ones(lo.shape, bool)
31         for i in range(3):
32             up = (tp*A[i] - bSM[i]*x).astype(np.float32)[None, :, None] - du[:, i][:, None, None]*y2[None]
33             dn = up - np.float32(tp); d = dc[:, i][:, None, None]
34             pos = d > 0
35             with np.errstate(divide='ignore', invalid='ignore'):
36                 lo = np.maximum(lo, np.where(pos, dn/d, -np.inf)); hi = np.minimum(hi, np.where(pos, up/d,
37                 ↪ np.inf))
38             ok &= pos | ((dn <= 0) & (up >= 0))
39             out[s:s+chunk] = (ok & (lo <= hi)).any(axis=(1, 2))
40     return out
41
42 if __name__ == "__main__":
43     ndec = int(sys.argv[1]) if len(sys.argv) > 1 else 20
44     X = lambda lam: math.log(lam/MZ); rng = np.random.default_rng(5)
45     t0 = time.time()
46     free = feasible2(A, X(1e10), X(1.22e19), 50)
47     print(f"Lambda free : {free.sum()} compatible spectra ({100*free.mean():.2f} %) [one scale:
48     ↪ 2565]", flush=True)
49     a, b = X(2.435e18), X(1.22e19)
50     pl = feasible2(A, a, b, 16)
51     print(f"Planck window : {pl.sum()} compatible spectra ({100*pl.mean():.2f} %) [one scale: 645]",
52     ↪ flush=True)
53     comp = S.sum(1); print("minimal complexity:", int(comp[pl].min()), "| time per evaluation ~", round((
54     ↪ time.time()-t0)/2), "s", flush=True)
55     for w in (1.25, 1.5):
56         dec = A*np.exp(rng.uniform(-math.log(w), math.log(w), (ndec, 3)))

```

```

51     nd = np.array([feasible2(d, a, b, 16).sum() for d in dec])
52     print(f"decoys x{w} ({ndec}): median {np.median(nd):.0f}, q90 {np.quantile(nd, .9):.0f}, P(decoy
    ↪ >= real) = {np.mean(nd >= pl.sum()):.2f}", flush=True)

```

B.9 09_architecture_comparison.py

Bayesian evidence, k_Y versus M_S , Higgs quartic (Sections 6.2–6.4).

```

1  #!/usr/bin/env python3
2  """
3  09 - Comparison of architectures with a penalty for continuous parameters (Bayesian evidence),
4      sensitivity of k_Y to an intermediate supersymmetry scale, and a cross-check with the Higgs.
5
6  A. Evidence Z = Int d(theta) prior(theta) L(theta), Laplace approximation.
7  Observables: (1/alpha_Y, 1/alpha_2, 1/alpha_3) at M_t. Gaussian likelihood, sigma = theoretical
8  error per coupling (unknown thresholds at M_U). Flat priors: ln M on [1 TeV, M_Pl],
9  1/alpha_U on [0, 100], k_Y (when free) on [1, 2]. The Occam factor automatically penalises
10 each continuous parameter; a discrete choice among n options costs a factor 1/n.
11 B. Required k_Y if supersymmetry appears at an intermediate scale M_S (SM below,
12    MSSM above), two loops.
13 C. Higgs quartic coupling lambda(mu) (approximate NLO): sign at M_U, and the Higgs mass
14    that lambda(M_U) >= 0 would require (supersymmetry broken at M_U).
15 """
16 import numpy as np, math
17 from fractions import Fraction as Fr
18 MT, MPL = 173.34, 1.22e19; K = 1/(16*math.pi**2); ainv = lambda g: 4*math.pi/g**2
19 g3_0 = 1.1666 + 0.00314*(0.1180-0.1184)/0.0007
20 Y0 = np.array([math.sqrt(5/3)*0.35830, 0.64779, g3_0, 0.93690]) # g1 (SU(5) norm.), g2, g3, y_t
    ↪ at M_t
21 OBS = np.array([(5/3)*ainv(Y0[0]), ainv(Y0[1]), ainv(Y0[2])])
22
23 MODELS = { # coefficients in SU(5) normalisation for g1
24     "SM": (np.array([41/10, -19/6, -7.0]), np.array([[199/50, 27/10, 44/5], [9/10, 35/6, 12.], [11/10, 9/2, -26.]]))
    ↪ ,
25     np.array([17/10, 3/2, 2.]), (4.5, 17/20, 9/4, 8.)),
26     "MSSM": (np.array([33/5, 1., -3.]), np.array([[199/25, 27/5, 88/5], [9/5, 25., 24.], [11/5, 9., 14.]]),
27             np.array([26/5, 6., 4.]), (6.0, 13/15, 3., 16/3)),
28 }
29 def beta(y, model, loops=2):
30     b, B, Ct, c = MODELS[model]; g, yt = y[:3], y[3]
31     dg = K*b*g**3 + (K*K*g**3*(B0(g**2) - Ct*yt**2) if loops == 2 else 0)
32     return np.append(dg, K*yt*(c[0]*yt**2 - c[1]*g[0]**2 - c[2]*g[1]**2 - c[3]*g[2]**2))
33 def run(y, t0, t1, model, n=600):
34     y, h = np.array(y, float), (t1-t0)/n
35     for _ in range(n):
36         k1 = beta(y, model); k2 = beta(y+h/2*k1, model); k3 = beta(y+h/2*k2, model); k4 = beta(y+h*k3,
    ↪ model)
37         y = y + h/6*(k1+2*k2+2*k3+k4)
38     return y
39 SB = math.sin(math.atan(10.)) # tan(beta) = 10
40 def up(tS, tU): # from M_t to M_U, SM then MSSM above M_S (tS=None: SM only)
41     if tS is None or tS >= tU: return run(Y0, math.log(MT), tU, "SM")
42     y = run(Y0, math.log(MT), tS, "SM"); y[3] /= SB
43     return run(y, tS, tU, "MSSM")
44 def down(tU, aUinv, kY, tS, ytU):
45     gU = math.sqrt(4*math.pi/aUinv); y = [math.sqrt(5/3)*gU/math.sqrt(kY), gU, gU, ytU]
46     if tS is None or tS >= tU: y = run(y, tU, math.log(MT), "SM")
47     else:
48         y = run(y, tU, tS, "MSSM"); y[3] *= SB; y = run(y, tS, math.log(MT), "SM")
49     return np.array([(5/3)*ainv(y[0]), ainv(y[1]), ainv(y[2])])
50
51 def crossing(tS):
52     """M_U where alpha_2 = alpha_3, 1/alpha_U, required k_Y."""
53     lo, hi = math.log(1e13), math.log(MPL)
54     for _ in range(40):
55         mid = (lo+hi)/2; y = up(tS, mid)
56         if ainv(y[1]) > ainv(y[2]): lo = mid
57         else: hi = mid
58     y = up(tS, mid); return mid, ainv(y[1]), (5/3)*ainv(y[0])/ainv(y[1]), y[3]
59
60 def evidence(kind, kY=None, tS=None, sigma=0.5):
61     tU, aU, kreq, ytU = crossing(tS)
62     if kind == "unif": # parameters (ln M_U, 1/alpha_U); k_Y fixed

```

```

63     th = np.array([tU, aU]); f = lambda p: down(p[0], p[1], kY, tS, ytU); vol = math.log(MPL/1e3)*100
64 elif kind == "kfree": # + continuous k_Y
65     th = np.array([tU, aU, kreq]); f = lambda p: down(p[0], p[1], p[2], tS, ytU); vol = math.log(MPL/1
    ↪ e3)*100*1.0
66 elif kind == "unif+MS": # + continuous ln M_S
67     th = np.array([tU, aU, tS]); f = lambda p: down(p[0], p[1], kY, p[2], ytU); vol = math.log(MPL/1e3
    ↪ )**2*100
68 elif kind == "het": # a single parameter g_s; M_s = 5.27e17 g_s
69     th = np.array([0.508]); f = lambda p: down(math.log(5.27e17*p[0]), 4*math.pi/p[0]**2, kY, None,
    ↪ ytU); vol = 1.0
70 eps = np.array([1e-3]*len(th))
71 for _ in range(6): # Gauss-Newton
72     r = f(th) - OBS
73     J = np.array([(f(th + e*np.eye(len(th))[j]) - OBS - r)/e for j, e in enumerate(eps)]).T
74     if J.shape[0] == J.shape[1]: th = th - np.linalg.solve(J, r)
75     else: th = th - np.linalg.lstsq(J, r, rcond=None)[0]
76 r = f(th) - OBS; chi2 = float(r@r)/sigma**2; n, m = 3, len(th)
77 lnZ = -0.5*chi2 + 0.5*(m-n)*math.log(2*math.pi*sigma**2) - 0.5*math.log(abs(np.linalg.det(J.T@J))) -
    ↪ math.log(vol)
78 return lnZ, chi2, th, r
79
80 if __name__ == "__main__":
81     print("== A. Evidence (Laplace approximation), two loops ==")
82     for sigma in (0.5, 1.0):
83         print(f" theoretical error sigma = {sigma} unit of 1/alpha per coupling")
84         rows = [("SM, k_Y = 5/3", "unif", 5/3, None), ("SM, k_Y = 4/3", "unif", 4/3, None),
85                 ("SM, k_Y = 13/10", "unif", 1.3, None), ("SM, k_Y = 32/25", "unif", 1.28, None),
86                 ("SM, k_Y free (continuous)", "kfree", None, None),
87                 ("MSSM at 3 TeV, k_Y = 5/3", "unif", 5/3, math.log(3e3)),
88                 ("MSSM, M_S free, k_Y = 5/3", "unif+MS", 5/3, math.log(3e3)),
89                 ("heterotic, k_Y = 4/3 (1 param.)", "het", 4/3, None)]
90         res = [(lab,)+evidence(kind, kY, tS, sigma) for lab, kind, kY, tS in rows]
91         ref = max(x[1] for x in res)
92         for lab, lnZ, chi2, th, r in res:
93             print(f" {lab:36s} chi2_min = {chi2:8.2f} ln Z = {lnZ:8.2f} Bayes factor / best = {math
    ↪ .exp(lnZ-ref):.2e}")
94     print("\n== B. Required k_Y versus the supersymmetry scale M_S (two loops) ==")
95     for MS in (1e3, 1e4, 1e6, 1e8, 1e10, 1e11, 1e12, 1e14, None):
96         tU, aU, kreq, _ = crossing(None if MS is None else math.log(MS))
97         print(f" M_S = {'%.0e GeV' % MS} if MS else 'none (pure SM)':>16s : M_U = {math.exp(tU):.2e}
    ↪ 1/alpha_U = {aU:5.2f} required k_Y = {kreq:.3f}")
98
99     print("\n== C. Higgs quartic (approximate NLO: lambda and y_t with leading two-loop terms) ==")
100     def lam_run(lam0, mt_shift=0.0):
101         y = np.append(Y0.copy(), lam0); y[3] += 0.00556*mt_shift
102         t, tU = math.log(MT), math.log(2.9e16); n = 4000; h = (tU-t)/n; tcross = None
103         def f(y):
104             g1, g2, g3, yt, l = y; gp2 = 0.6*g1*g1
105             dg = beta(y[:4], "SM")
106             dyt2 = K*K*yt*(-12*yt**4 + 36*g3*g3*yt*yt - 108*g3**4 + 6*l*1 - 12*l*yt*yt)
107             dl = K*(24*l*1 + 12*l*yt*yt - 6*yt**4 - 3*l*(3*g2*g2+gp2) + 0.375*(2*g2**4 + (g2*g2+gp2)**2))
    ↪ \
108                 + K*K*(-312*l**3 - 144*l*1*yt*yt - 3*l*yt**4 + 30*yt**6 + 80*l*g3*g3*yt*yt - 32*g3*g3*yt
    ↪ **4)
109             dg[3] += dyt2; return np.append(dg, dl)
110         for i in range(n):
111             k1 = f(y); k2 = f(y+h/2*k1); k3 = f(y+h/2*k2); k4 = f(y+h*k3); yn = y + h/6*(k1+2*k2+2*k3+k4)
112             if tcross is None and y[4] > 0 >= yn[4]: tcross = t + i*h
113             y = yn
114         return y[4], tcross
115     for dmt in (0.0, -0.64, -2.0):
116         lU, tc = lam_run(0.12604, dmt)
117         print(f" M_t = {173.34+dmt:6.2f} GeV : lambda(M_U) = {lU:+.4f} ; lambda = 0 at mu = " + (f"{
    ↪ math.exp(tc):.1e} GeV" if tc else "never"))
118     lo, hi = 0.12, 0.16
119     for _ in range(30):
120         mid = (lo+hi)/2
121         if lam_run(mid)[0] < 0: lo = mid
122         else: hi = mid
123     print(f" lambda(M_U) >= 0 requires lambda(M_t) >= {mid:.4f}, i.e. M_h >~ {125.15 + (mid-0.12604)
    ↪ *125.15/(2*0.12604):.1f} GeV (measured: 125.2; NNLO literature: ~129-130)")
124     g2U = 4*math.pi/46.18
125     print(f" supersymmetric window at M_U: 0 <= lambda <= (g2^2+gY^2)/8 = {(g2U + g2U*3/4)/8:.4f}")
126
127     # == D. Evidence of the 'rigid strong coupling' family (script 03), one loop ==

```

```

128 # Likelihood: c_i = alpha_i^-1(Lambda) uniform in [0,1] (density 1 per unit of 1/alpha),
129 # flat priors in ln(Lambda) and ln(M) on [1 TeV, M_Pl], uniform prior over the 151,263 spectra.
130 import itertools
131 tp = 2*math.pi; MZ = 91.1876
132 A = np.array([127.951*(1-0.23122), 127.951*0.23122, 1/0.1180]); bSM = np.array([41/6, -19/6, -7.0])
133 C = np.array([(4/3,0,0),(2/3,2/3,0),(4/9,0,2/3),(16/9,0,2/3),(2/9,2,4/3),(0,4/3,0),(0,0,2)])
134 S = np.array(list(itertools.product(*[range(7)]*5, range(3), range(3)))); db = S@C
135 x = np.linspace(math.log(1e10/MZ), math.log(MPL/MZ), 200); dx = x[1]-x[0]; ymax = x - math.log(1e3/MZ)
136 lo = np.zeros((len(S), len(x))); hi = np.broadcast_to(ymax, lo.shape).copy(); ok = np.ones(lo.shape,
    ↪ bool)
137 for i in range(3):
138     upb = tp*A[i] - bSM[i]*x; dn = upb - tp; d = db[:, i][:, None]
139     with np.errstate(divide='ignore', invalid='ignore'):
140         lo = np.maximum(lo, np.where(d > 0, dn/d, -np.inf)); hi = np.minimum(hi, np.where(d > 0, upb/d
    ↪ , np.inf))
141     ok &= np.where(d > 0, True, (dn <= 0) & (upb >= 0))
142 area = (np.where(ok, np.clip(hi-lo, 0, None), 0).sum(1)*dx)
143 V = math.log(MPL/1e3)**2
144 print("\n== D. 'Rigid strong coupling' family (one loop) ==")
145 print(f" ln Z (average over spectra) = {math.log(area.mean()/V):.2f}; best single spectrum: ln Z =
    ↪ {math.log(area.max()/V):.2f}")

```